# SAGAR PUBLIC SCHOOL

**Integrated Programme | Topic: Gravitation**

## PRACTICE SHEET 2

### Gravitational Potential Energy, Potential, Escape Velocity and Orbits

**Time Allowed: 60 Mins** **Total Questions: 18**

Name: _____ Class/Sec: _____ Date: _____

**Instructions:** Attempt all questions. Questions in the *one or more than one option is correct* category are marked accordingly. Options are labelled as in the source book (A–D or 1–4). The answer key and a self-assessment page follow the questions.

**Q.1** Two masses $m_1$ & $m_2$ are initially at rest and are separated by a very large distance. If the masses approach each other subsequently, due to gravitational attraction between them, their relative velocity of approach at a separation distance of $d$ is

(A) $\dfrac{2Gd}{(m_1+m_2)}$ (B) $\dfrac{(m_1+m_2)G}{2d}$

(C) $\left[(m_1+m_2)\dfrac{2G}{d}\right]^{1/2}$ (D) $(m_1+m_2)^{1/2}\,2Gd$

**Q.2** A spherical uniform planet is rotating about its axis. The velocity of a point on its equator is $V$. Due to the rotation of planet about its axis the acceleration due to gravity $g$ at equator is $1/2$ of $g$ at poles. The escape velocity of a particle on the pole of planet in terms of $V$.

(A) $V_e = 2V$ (B) $V_e = V$ (C) $V_e = V/2$ (D) $V_e = \sqrt{3}\,V$

**Q.3** Two planets A and B have the same material density. If the radius of A is twice that of B, then the ratio of the escape velocity $\dfrac{v_A}{v_B}$ is

(A) 2 (B) $\sqrt{2}$ (C) $1/\sqrt{2}$ (D) $1/2$

**Q.4** A hollow spherical shell is compressed to half its radius. The gravitational potential at the centre

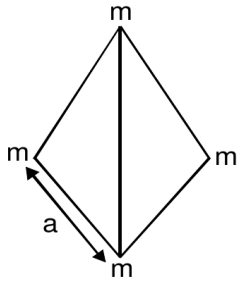
(A) increases (B) decreases

(C) remains same (D) during the compression increases then returns at the previous value

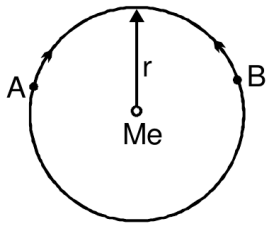
**Q.5** *(One or more than one option is correct)* Two masses $m_1$ and $m_2$ $(m_1 < m_2)$ are released from rest from a finite distance. They start under their mutual gravitational attraction

(A) acceleration of $m_1$ is more than that of $m_2$
(B) acceleration of $m_2$ is more than that of $m_1$
(C) centre of mass of system will remain at rest in all the references frame
(D) total energy of system remains constant

**Q.6** Four masses (each of $m$) are placed at the vertices of a regular pyramid (triangular base) of side $a$. Find the work done by the system while taking them apart so that they form the pyramid of side $2a$.



**Q.7** Consider two satellites A and B of equal mass $m$, moving in the same circular orbit of radius $r$ around the earth E but in opposite sense of rotation and therefore on a collision course (see figure).



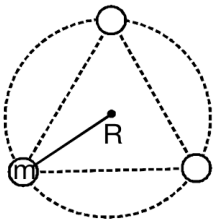

(a) In terms of $G$, $M_e$, $m$ and $r$ find the total mechanical energy $E_A + E_B$ of the two satellite plus earth system before collision.

(b) If the collision is completely inelastic so that wreckage remains as one piece of tangled material (mass $= 2m$), find the total mechanical energy immediately after collision.

(c) Describe the subsequent motion of the wreckage.

**Q.8** Find the potential energy of a system of eight particles placed at the vertices of a cube of side $L$. Neglect the self energy of the particles.

**Q.9** A hypothetical planet of mass $M$ has three moons each of equal mass $m$ each revolving in the same circular orbit of radius $R$. The masses are equally spaced and thus form an equilateral triangle. Find




(i) the total P.E. of the system

(ii) the orbital speed of each moon such that they maintain this configuration.

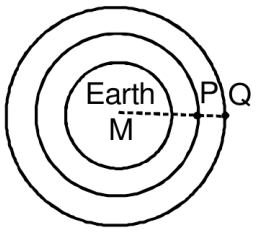
**Q.10** A body moving radially away from a planet of mass $M$, when at distance $r$ from planet, explodes in such a way that two of its many fragments move in mutually perpendicular circular orbits around the planet. What will be

(a) the velocity in circular orbits.

(b) maximum distance between the two fragments before collision and

(c) magnitude of their relative velocity just before they collide.

**Q.11** A satellite P is revolving around the earth at a height $h$ = radius of earth ($R$) above equator. Another satellite Q is at a height $2h$ revolving in opposite direction. At an instant the two are at same vertical line passing through centre of earth. Find the least time after which again they are in this situation.




**Q.12** A ring of radius $R$ is made from a thin wire of radius $r$. If $\rho$ is the density of the material of wire then what will be the gravitational force exerted by the ring on the material particle of mass $m$ placed on the axis of ring at a distance $x$ from its centre. Show that the force will be maximum when $x = R/\sqrt{2}$ and the maximum value of force will be given as

$$F_{\max} = \frac{4\pi^2 G r^2 \rho\, m}{(3)^{3/2} R}$$

**Q.13** A satellite of mass $m$ is in an elliptical orbit around the earth of mass $M$ ($M \gg m$). The speed of the satellite at its nearest point to the earth (perigee) is $\sqrt{\dfrac{6GM}{5R}}$ where $R$ = its closest distance to the earth. It is desired to transfer this satellite into a circular orbit around the earth of radius equal its largest distance from the earth. Find the increase in its speed to be imparted at the apogee (farthest point on the elliptical orbit).

**Q.14** A small body of mass $m$ is projected with a velocity just sufficient to make it reach from the surface of a planet (of radius $2R$ and mass $3M$) to the surface of another planet (of radius $R$ and mass $M$). The distance between the centers of the two spherical planets is $6R$. The distance of the body from the center of bigger planet is $x$ at any moment. During the journey, find the distance $x$ where the speed of the body is (a) maximum (b) minimum. Assume motion of body along the line joining centres of planets.

**Q.15** Two uniform spherical stars made of same material have radii $R$ and $2R$. Mass of the smaller planet is $m$. They start moving from rest towards each other from a large distance under mutual force of gravity. The collision between the stars is inelastic with coefficient of restitution 1/2.

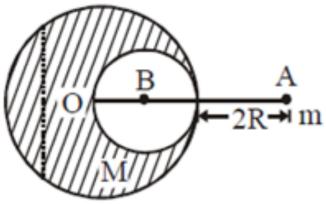
(a) Find the kinetic energy of the system just after the collision.

(b) Find the maximum separation between their centres after their first collision.

**Q.16** Two identical satellites are at the heights $R$ and $7R$ from the Earth's surface. Then which of the following statement is **incorrect**. ($R$ = radius of the Earth)

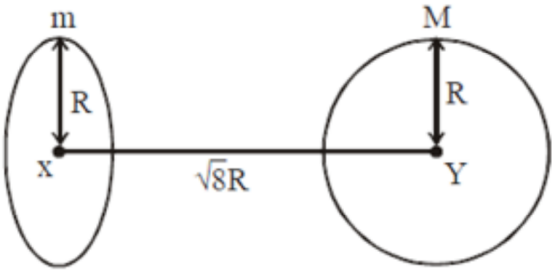
(A) Ratio of total energy of both is 5
(B) Ratio of kinetic energy of both is 4
(C) Ratio of potential energy of both 4
(D) Ratio of total energy of both is 4 and ratio of magnitude of potential to kinetic energy is 2

**Q.17** A solid sphere of radius $R$ gravitationally attracts a particle placed at $3R$ from its centre with a force $F_1$. Now a spherical cavity of radius $\left(\frac{R}{2}\right)$ is made in the sphere (as shown in figure) and the force becomes $F_2$. The value of $F_1 : F_2$ is: **[JEE Main-2021]**




(1) 25 : 36 (2) 36 : 25 (3) 50 : 41 (4) 41 : 50

**Q.18** Find the gravitational force of attraction between the ring and sphere as shown in the diagram, where the plane of the ring is perpendicular to the line joining the centres. If $\sqrt{8}\,R$ is the distance between the centres of a ring (of mass $m$) and a sphere (mass $M$) where both have equal radius $R$. **[JEE Main-2021]**




(1) $\frac{\sqrt{8}}{9} \cdot \frac{GmM}{R}$ (2) $\frac{2\sqrt{2}}{3} \cdot \frac{GmM}{R^2}$ (3) $\frac{1}{3\sqrt{8}} \cdot \frac{GmM}{R^2}$ (4) $\frac{\sqrt{8}}{27} \cdot \frac{GmM}{R^2}$

# ANSWER KEY – PRACTICE SHEET 2

Answers are taken from the answer keys supplied with the source books (**Motion**: Answer_key.pdf; **Nucleus**: Answer Key at the end of GRAVITATION.pdf).

| Q. No. | Source | Answer |
|---|---|---|
| 1 | Motion Ex-I Q.2 | (C) |
| 2 | Motion Ex-I Q.8 | (A) |
| 3 | Motion Ex-I Q.9 | (A) |
| 4 | Motion Ex-I Q.11 | (B) |
| 5 | Motion Ex-II Q.2 | (A), (D) |
| 6 | Motion Ex-III Q.1 | $-\dfrac{3Gm^2}{a}$ |
| 7 | Motion Ex-III Q.9 | (a) $-\dfrac{GmM_e}{r}$ (b) $-\dfrac{2GmM_e}{r}$ (c) not given in the source key |
| 8 | Motion Ex-III Q.12 | $-\dfrac{4GM^2}{L}\left[3+\dfrac{3}{\sqrt{2}}+\dfrac{1}{\sqrt{3}}\right]$ ($M$ = mass of each particle, as used in the source key) |
| 9 | Motion Ex-III Q.13 | (i) $-\dfrac{3Gm}{R}\left(\dfrac{m}{\sqrt{3}}+M\right)$ (ii) $\sqrt{\dfrac{G}{R}\left(\dfrac{m}{\sqrt{3}}+M\right)}$ |
| 10 | Motion Ex-III Q.16 | (a) $\sqrt{\dfrac{GM}{r}}$ (b) $r\sqrt{2}$ (c) $\sqrt{\dfrac{2GM}{r}}$ |
| 11 | Motion Ex-IV Q.1 | $\dfrac{2\pi R^{3/2}\,(6\sqrt{6})}{\sqrt{GM}\,(2\sqrt{2}+3\sqrt{3})}$ |
| 12 | Motion Ex-IV Q.5 | "Show that question – the result is stated in the question; no separate answer is given in the source key |
| 13 | Motion Ex-IV Q.8 | $\sqrt{\dfrac{GM}{R}}\left[\sqrt{\dfrac{2}{3}}-\sqrt{\dfrac{8}{15}}\right]$ |
| 14 | Nucleus S-1 Q.2 | (a) $2R$ (b) $3R\,(3-\sqrt{3})$ |
| 15 | Nucleus S-1 Q.3 | (a) $\dfrac{2Gm^2}{3R}$ (b) $4R$ |
| 16 | Nucleus O-1 Q.14 | (A) |
| 17 | Nucleus JM Q.36 | (3) |
| 18 | Nucleus JM Q.39 | (4) |

**Source abbreviations:** Ex-I/II/III/IV = Exercise I/II/III/IV of the Motion booklet; O-1, S-1, JM = Exercise (O-1), (S-1), (JM) of the Nucleus booklet.

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## SELF-ASSESSMENT – PRACTICE SHEET 2

| | |
|---|---|
| **Time Allowed:** 60 Mins | **Time Taken:** _____ Mins |
| **Total Questions:** 18 | **Questions Attempted:** _____ |
| **Correct:** _____ | **Incorrect:** _____ |
| **Unattempted:** _____ | **Accuracy:** _____% (Correct / Attempted ×100) |

**Question-wise record** (tick one: C = Correct, W = Wrong, S = Skipped)

| | | | |
|---|---|---|---|
| **Q.1** ☐C ☐W ☐S | **Q.2** ☐C ☐W ☐S | **Q.3** ☐C ☐W ☐S | **Q.4** ☐C ☐W ☐S |
| **Q.5** ☐C ☐W ☐S | **Q.6** ☐C ☐W ☐S | **Q.7** ☐C ☐W ☐S | **Q.8** ☐C ☐W ☐S |
| **Q.9** ☐C ☐W ☐S | **Q.10** ☐C ☐W ☐S | **Q.11** ☐C ☐W ☐S | **Q.12** ☐C ☐W ☐S |
| **Q.13** ☐C ☐W ☐S | **Q.14** ☐C ☐W ☐S | **Q.15** ☐C ☐W ☐S | **Q.16** ☐C ☐W ☐S |
| **Q.17** ☐C ☐W ☐S | **Q.18** ☐C ☐W ☐S | | |

**Reason for mistakes** (tick all that apply)

☐ Concept not clear ☐ Formula forgotten ☐ Calculation mistake

☐ Misread the question ☐ Ran out of time ☐ Guess work

**Personal Feedback**

*What went well:*

_____

_____

*Topics / questions I need to revise:*

_____

_____

*My action plan before the next sheet:*

_____

_____

*Teacher's remarks:*

_____

_____

Student's signature: _____ Teacher's signature: _____